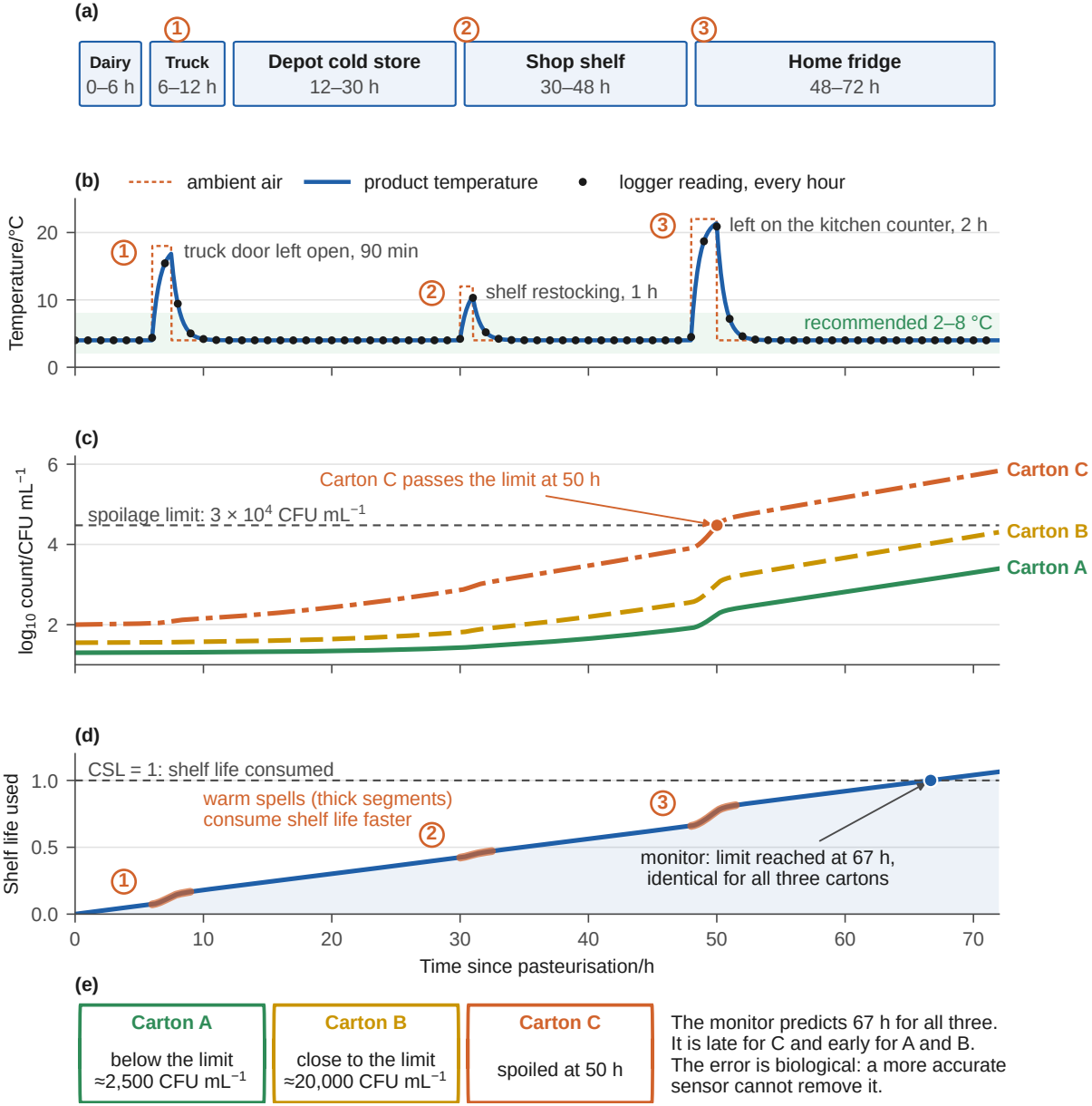

(a)
1
Dairy
0–6 h
Truck
6–12 h
2
Depot cold store
12–30 h
Shop shelf
30–48 h
3
Home fridge
48–72 h
(b)
ambient air
product temperature
logger reading, every hour
Temperature/°C
20
10
0
1
truck door left open, 90 min
2
shelf restocking, 1 h
3
left on the kitchen counter, 2 h
recommended 2–8 °C
(c)
log10 count/CFU mL−1
6
4
2
Carton C passes the limit at 50 h
spoilage limit: 3 × 10^4 CFU mL−1
Carton C
Carton B
Carton A
(d)
Shelf life used
1.0
0.5
0.0
CSL = 1: shelf life consumed
warm spells (thick segments)
consume shelf life faster
1
2
3
monitor: limit reached at 67 h,
identical for all three cartons
0
10
20
30
40
50
60
70
Time since pasteurisation/h
(e)
Carton A
below the limit
≈2,500 CFU mL−1
Carton B
close to the limit
≈20,000 CFU mL−1
Carton C
spoiled at 50 h
The monitor predicts 67 h for all three.
It is late for C and early for A and B.
The error is biological: a more accurate
sensor cannot remove it.